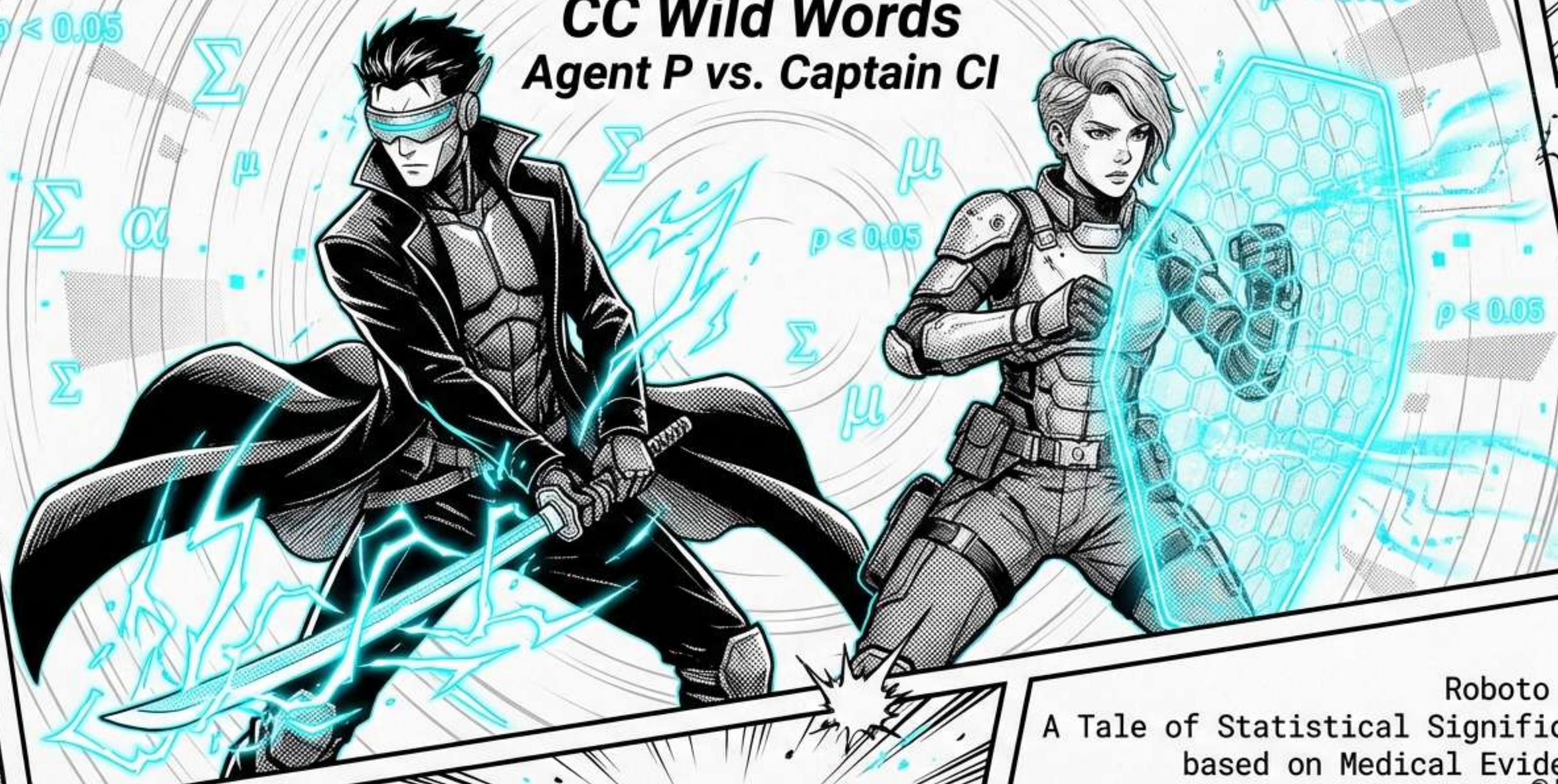


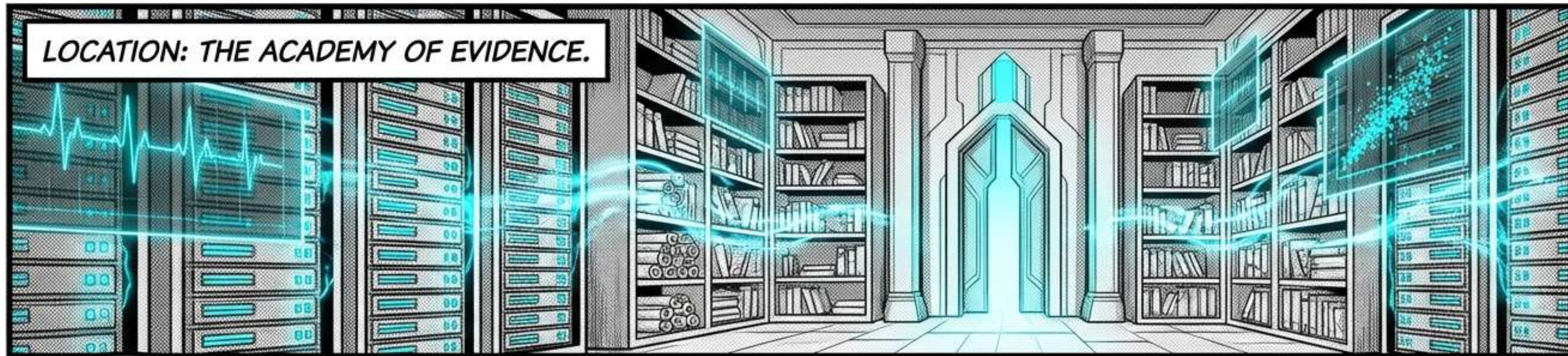
Komika Axis **THE DUEL OF INFERENCE**

CC Wild Words
Agent P vs. Captain CI

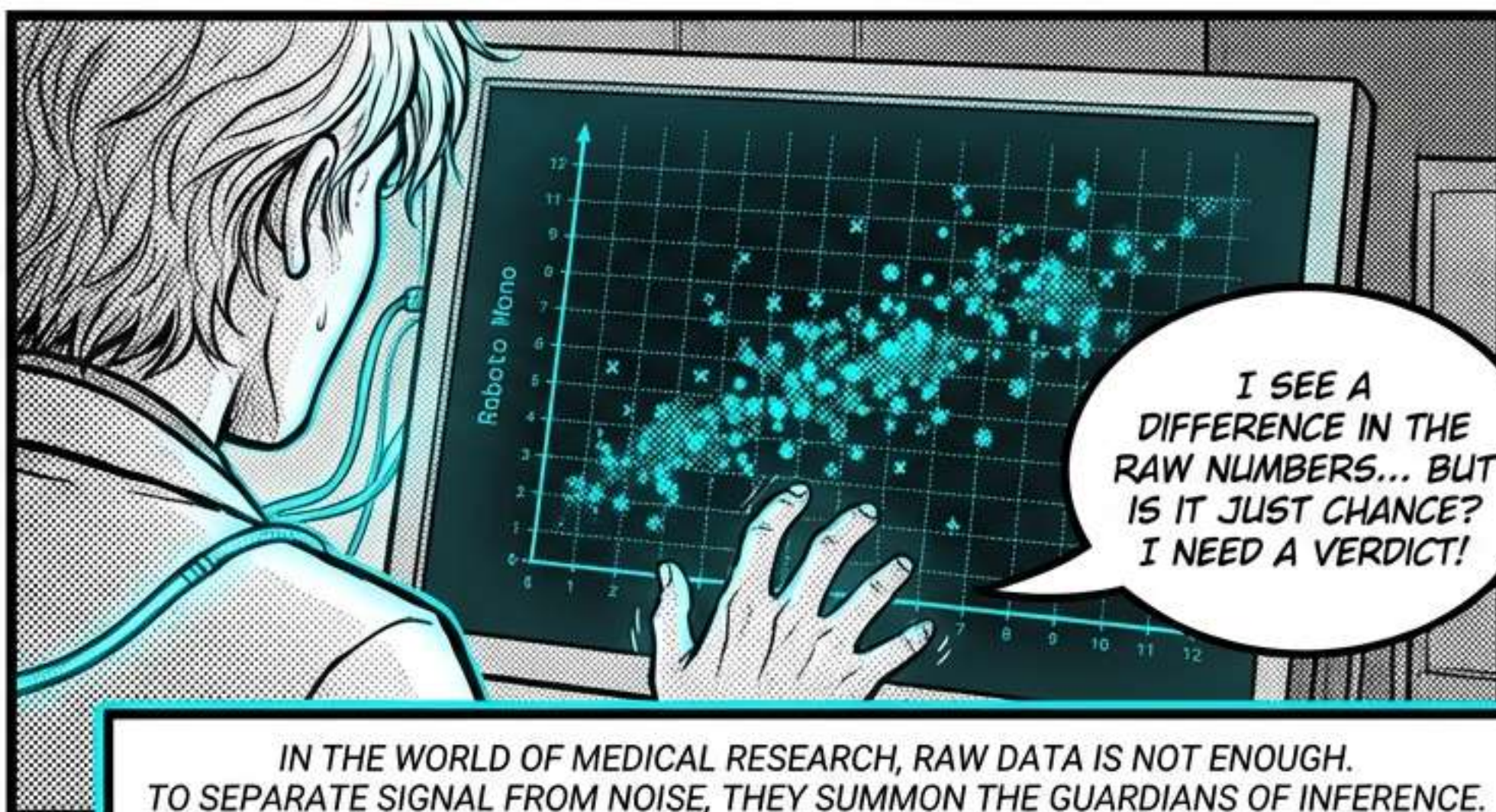
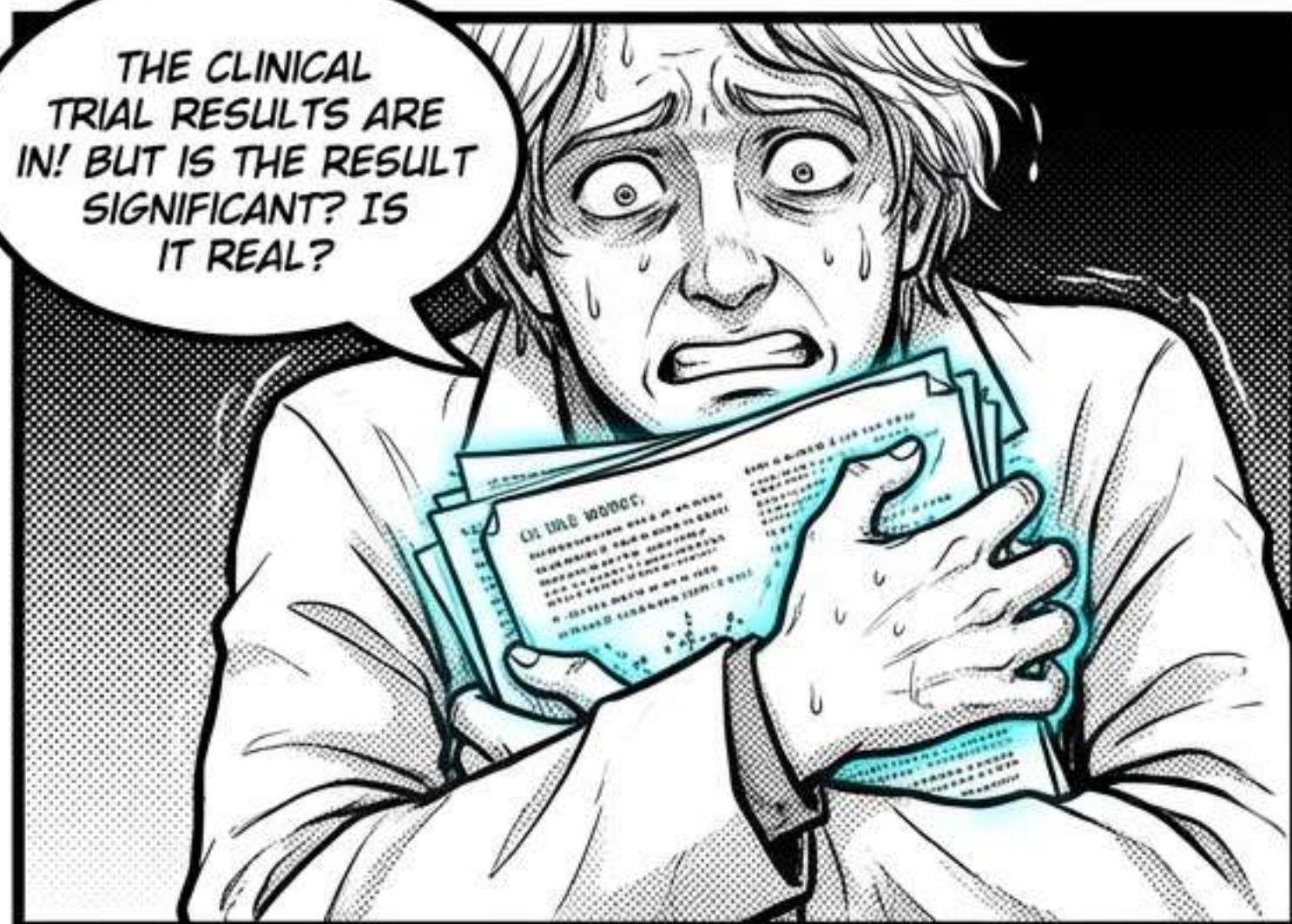


Roboto Mono
A Tale of Statistical Significance
based on Medical Evidence.

LOCATION: THE ACADEMY OF EVIDENCE.



THE CLINICAL
TRIAL RESULTS ARE
IN! BUT IS THE RESULT
SIGNIFICANT? IS
IT REAL?



IN THE WORLD OF MEDICAL RESEARCH, RAW DATA IS NOT ENOUGH.
TO SEPARATE SIGNAL FROM NOISE, THEY SUMMON THE GUARDIANS OF INFERENCE.

The Enemy: The Null Hypothesis (H_0).

H_0

I am the *status quo*.
I am "No Association."
I am "No Difference."

Technical Brief: H_0 represents the statement that there is no relationship between variables.

Example: "There is no difference in viral load reduction between the two antiretroviral drugs."

AGENT P

(The P-Value)

Tech Hud Overlay

Power: The Blade of Significance.

POWER DEFINITION

The Probability of obtaining a result as extreme or more extreme as found in the study, if the Null Hypothesis were true.

*I don't judge if the hypothesis is true. I judge the **EVIDENCE** against it. How unlikely is this data?*

The Threshold of Significance.
Conventionally set at 5%.

You are
unlikely to exist
by chance alone!
Rejected!

NOT SIGNIFICANT
($P > 0.05$)

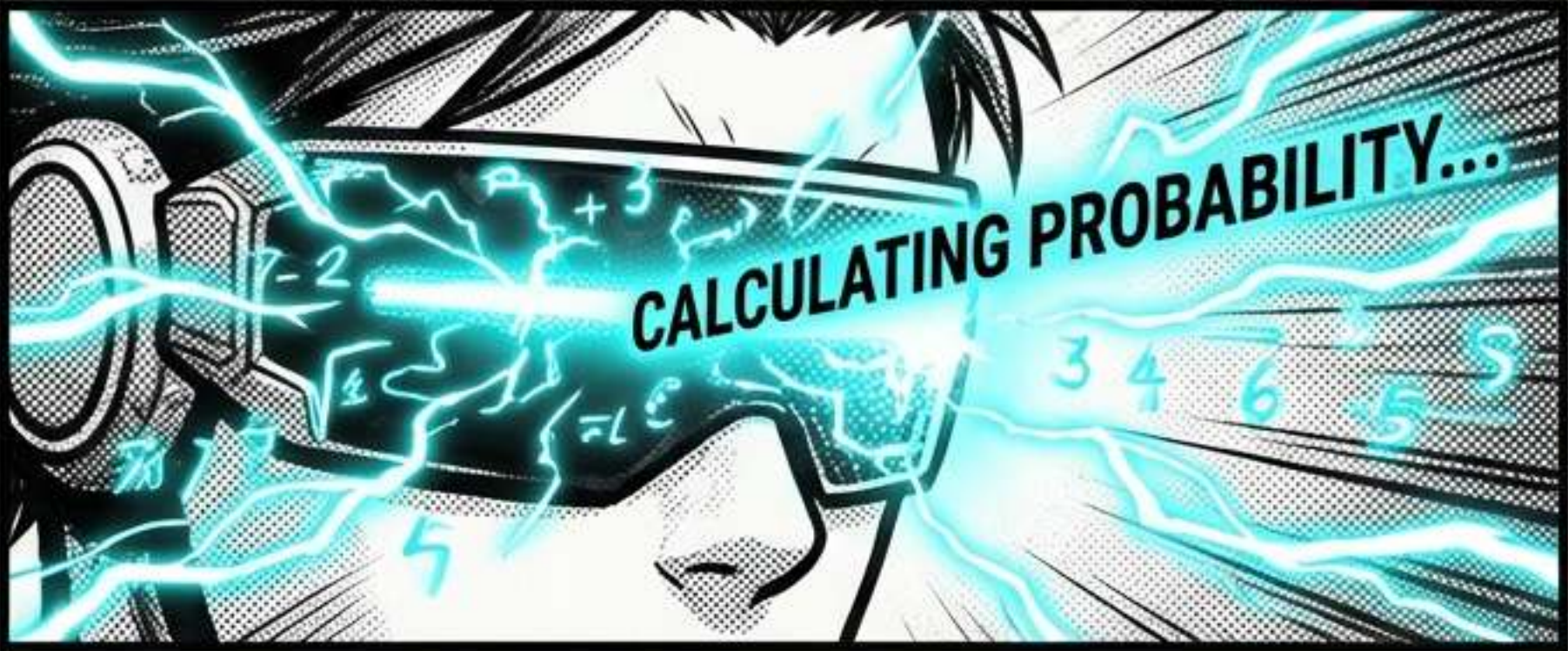
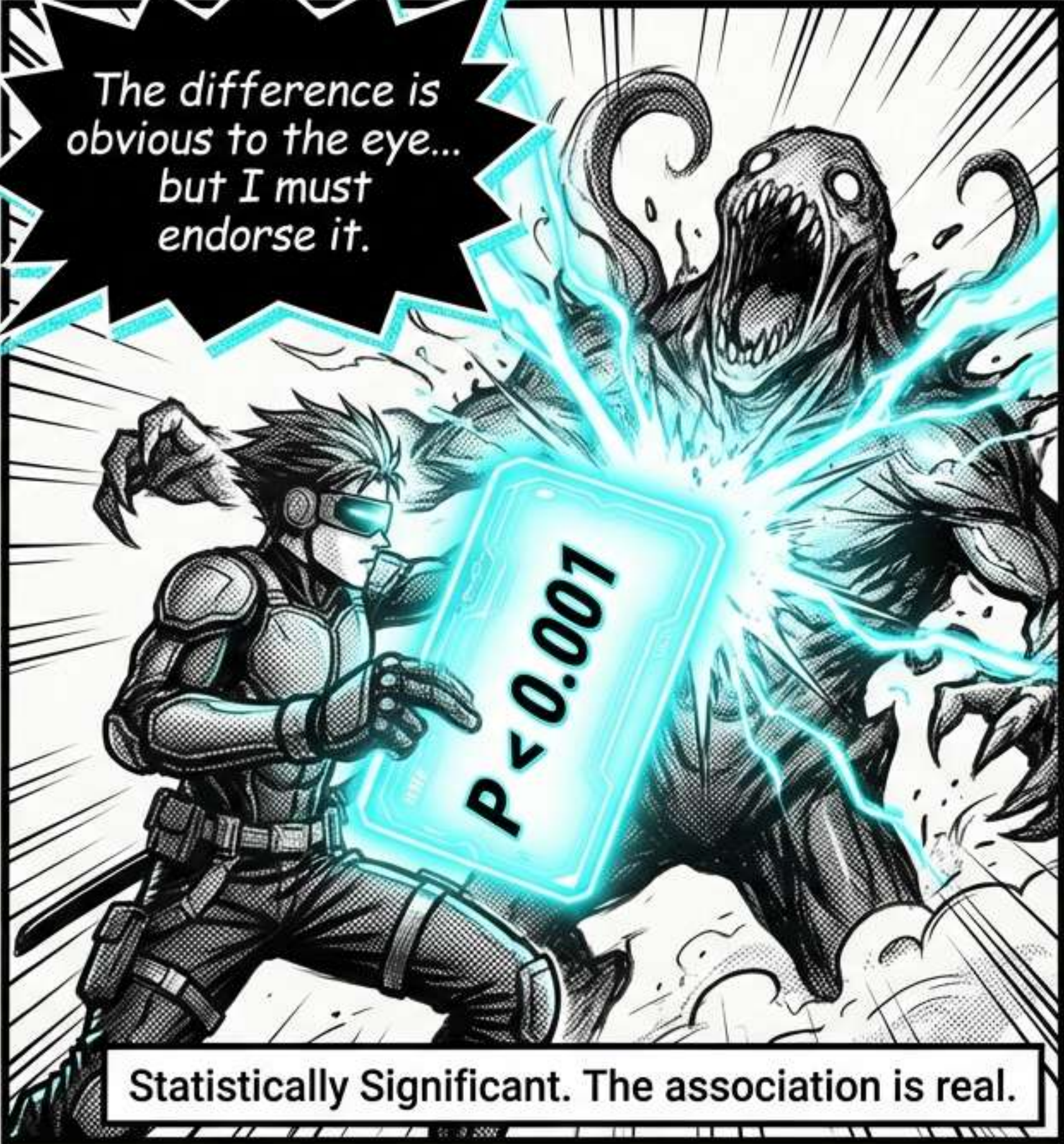
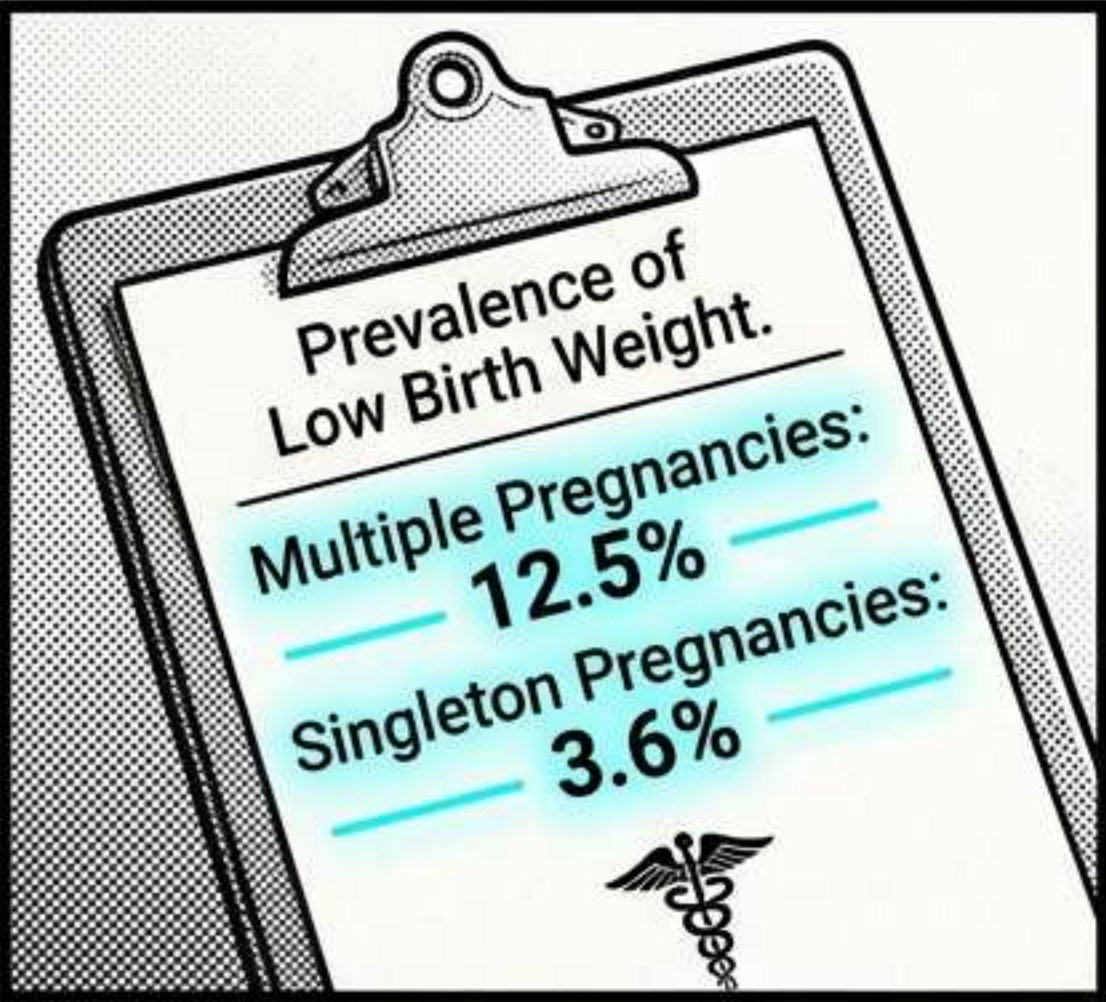
— $\alpha = 0.05$

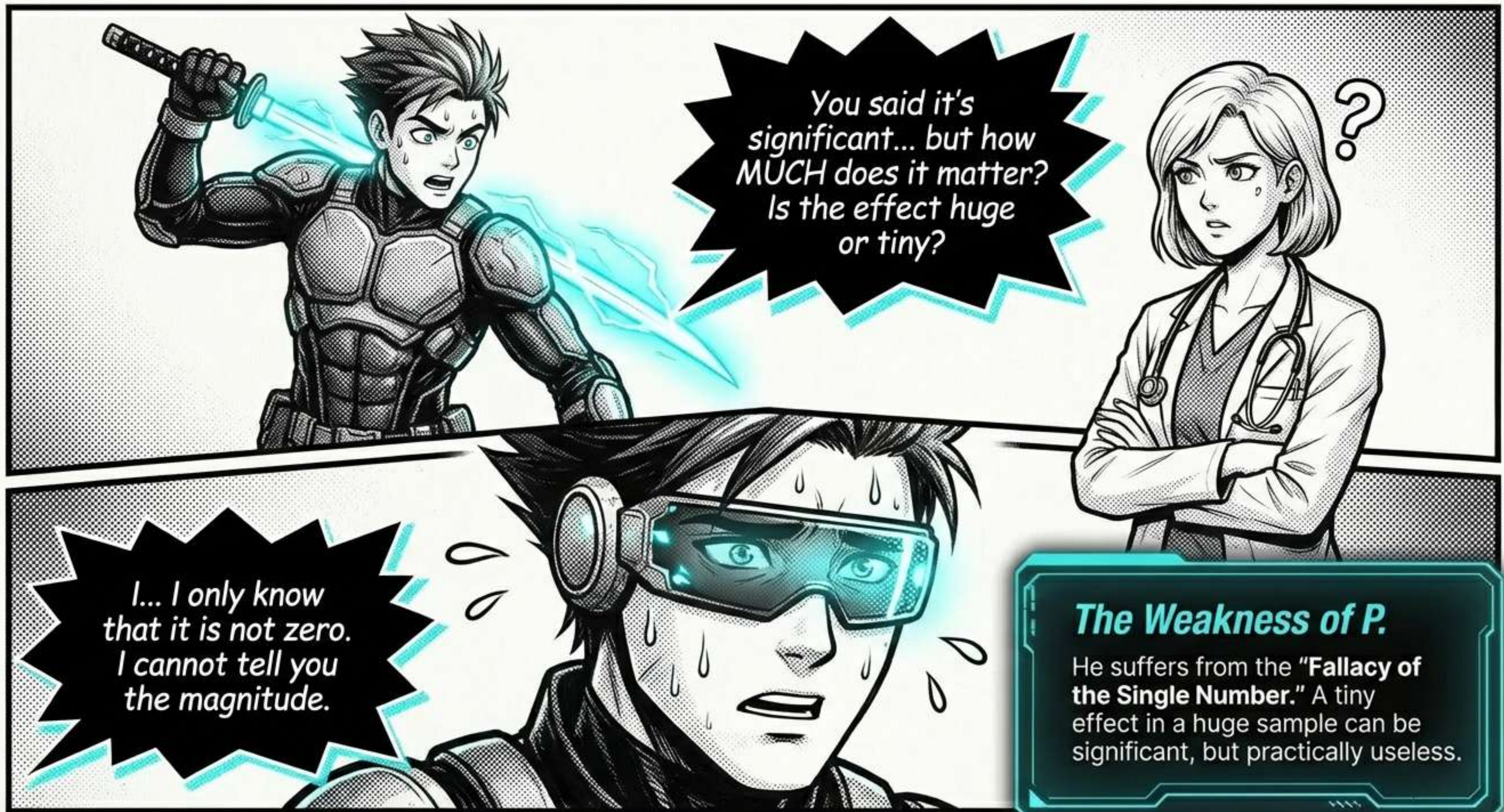
SIGNIFICANT
($P < 0.05$)

SLASH!

If $P < 0.05$: The evidence against
the Null is strong. We reject H_0 .

BATTLE CASE STUDY: LOW BIRTH WEIGHT





You said it's significant... but how **MUCH** does it matter? Is the effect huge or tiny?

I... I only know that it is not zero. I cannot tell you the magnitude.

The Weakness of P.

He suffers from the "**Fallacy of the Single Number.**" A tiny effect in a huge sample can be significant, but practically useless.

CAPTAIN CI (The Confidence Interval)

POWER DEFINITION:

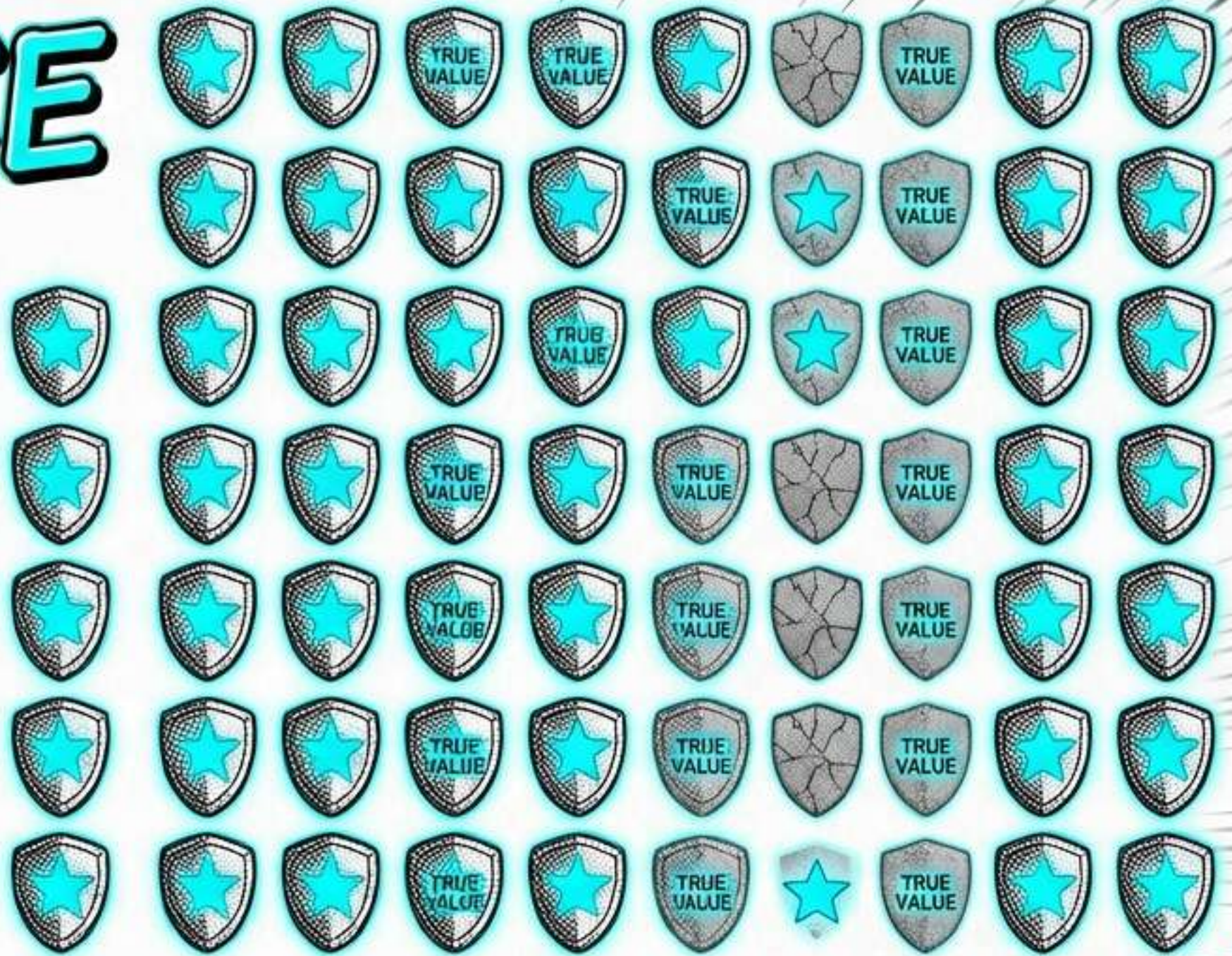
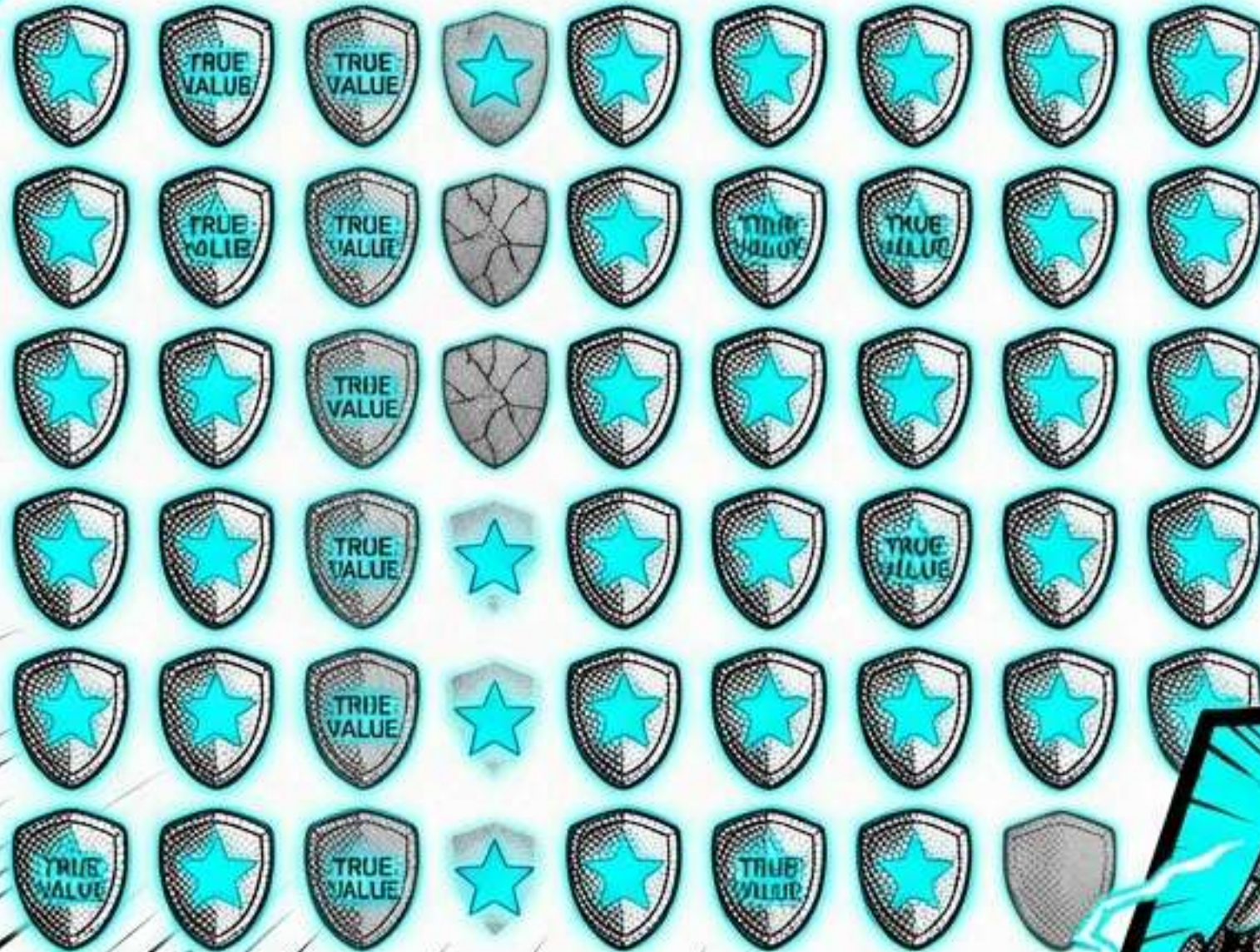
A range of values within which we are reasonably confident (usually 95%) that the true population parameter lies.

Power:
The Shield of Estimation
(95% CI).

Stand back, P.
You provide a verdict.
I provide CONTEXT.
(CC Wild Words)

CONFIDENCE

Imagine the study is repeated 100 times.



I do not promise certainty for this single trial. I promise that my METHOD captures the truth 95% of the time.

Scenario:
Reduction in Blood Sugar
(Difference in Means).

**RESULT:
NOT SIGNIFICANT.**

[-5 to +15]

My shield includes Zero!
It is plausible there
is no difference.

Sidebar Note:
Null Value for
Difference = 0.
Roboto Mono

Scenario:
Alcohol Intake vs. Liver
Disease (Odds Ratio).

My shield excludes
the Null Value of 1!
The risk is real.

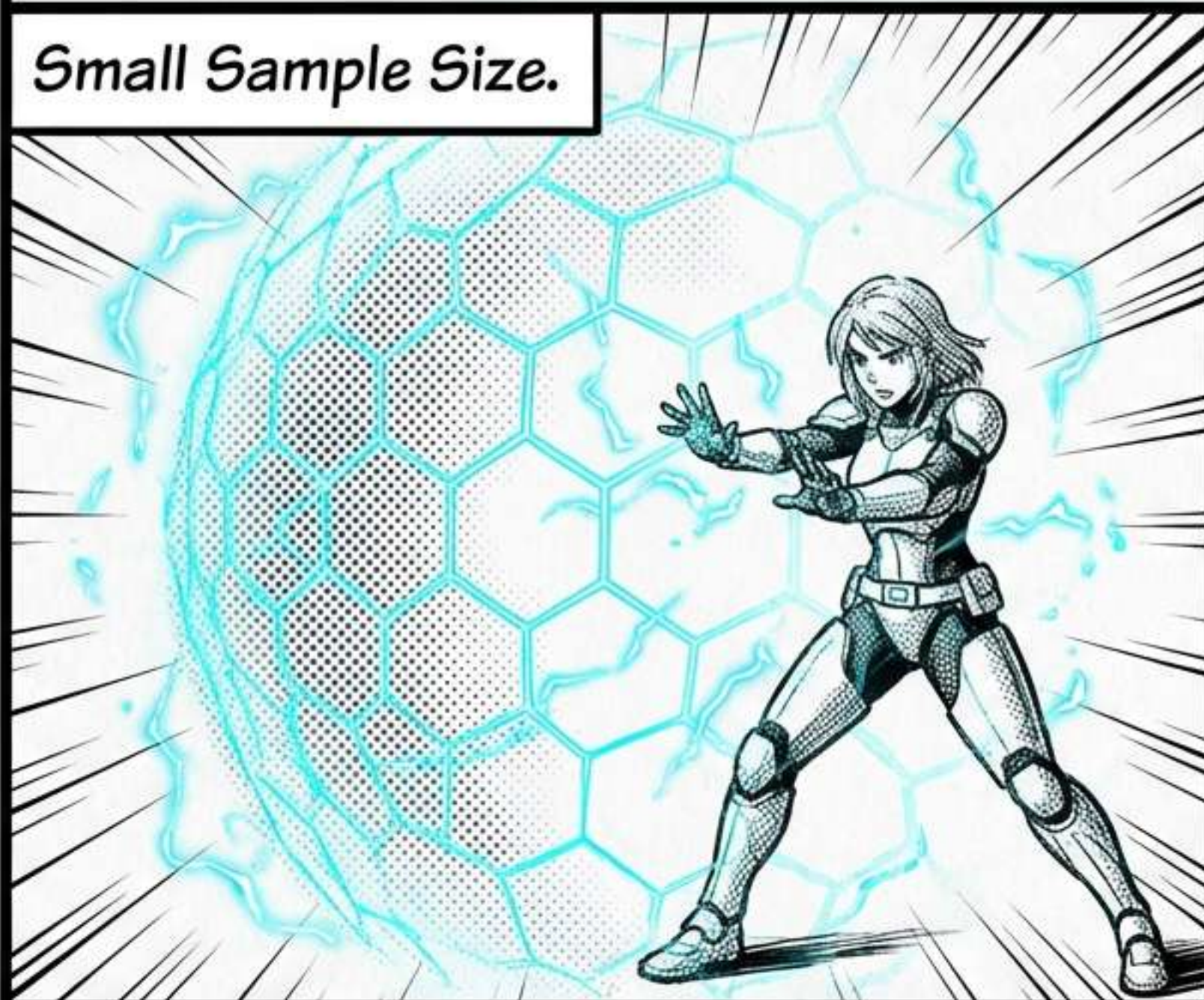
**RESULT:
SIGNIFICANT!**

[1.5 to 4.2]

Sidebar Note:
Null Value for
Ratio = 1.
Roboto Mono

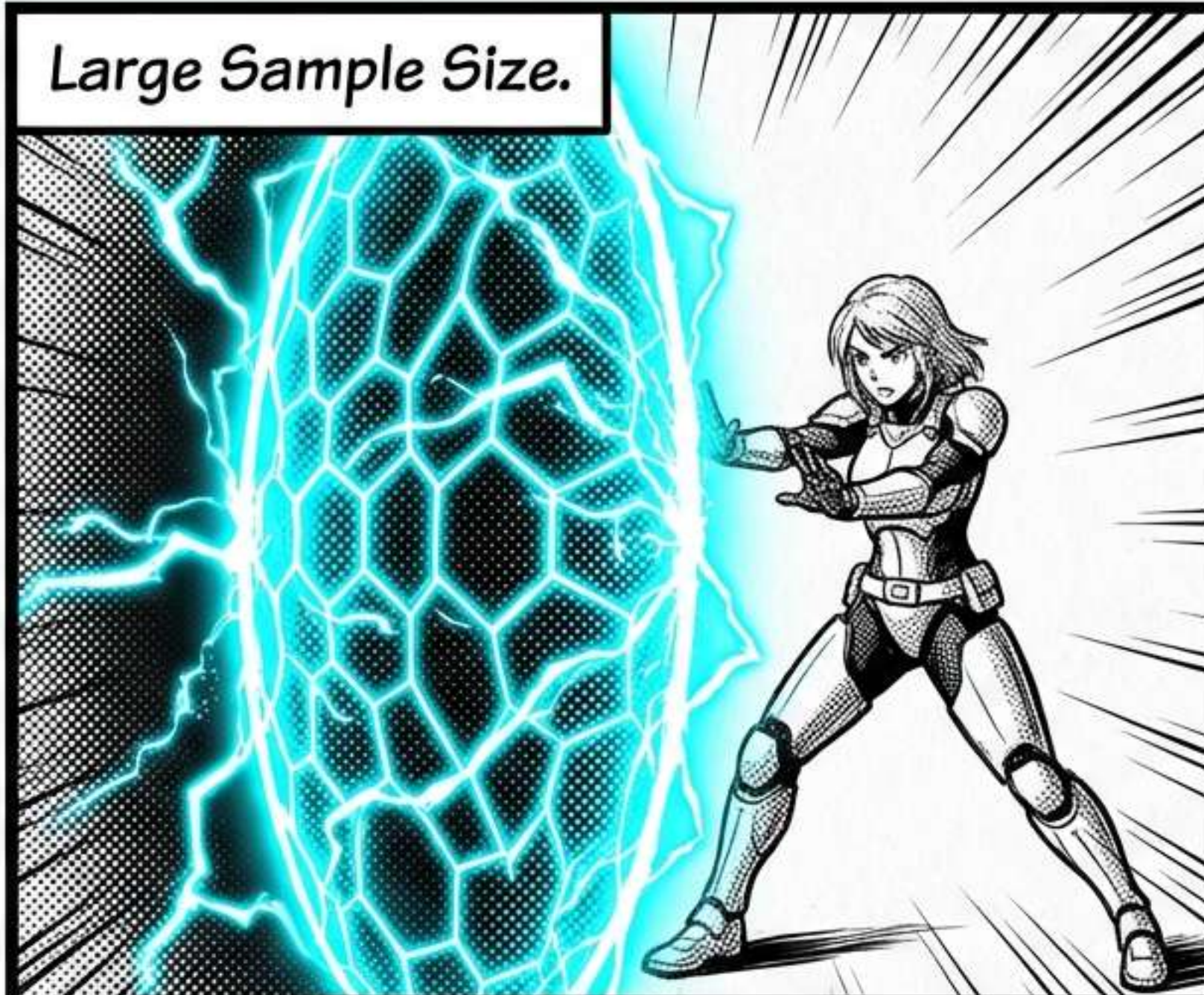
Captain CI reveals the **RELIABILITY** of the estimate.
More data equals a sharper shield.

Small Sample Size.

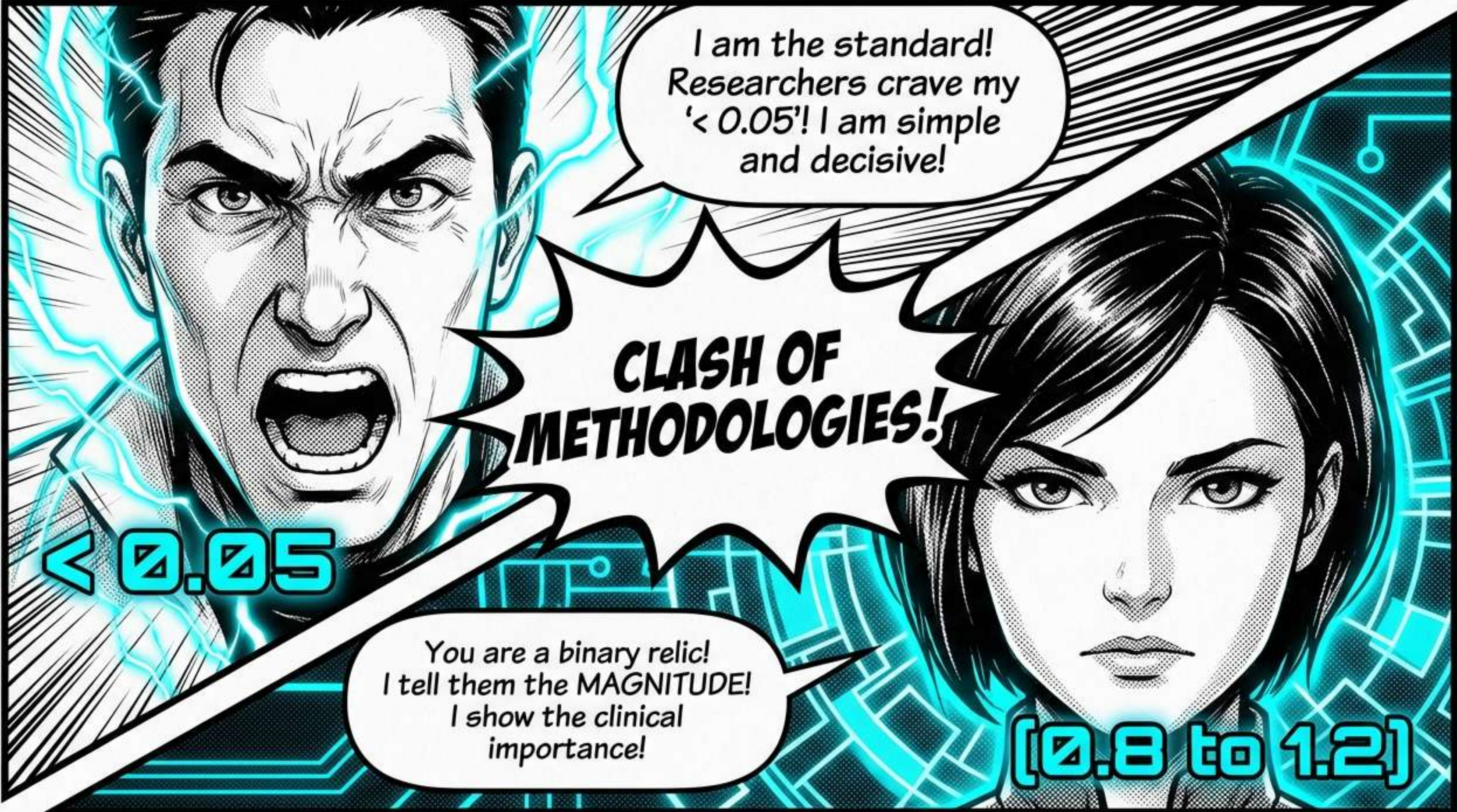


Wide Interval = High Uncertainty.

Large Sample Size.



Narrow Interval = High Precision.



I am the standard!
Researchers crave my
' < 0.05 '! I am simple
and decisive!

CLASH OF METHODOLOGIES!

< 0.05

You are a binary relic!
I tell them the **MAGNITUDE!**
I show the clinical
importance!

[0.8 to 1.2]

TECHNICAL SPECS COMPARISON

Power Level Analysis

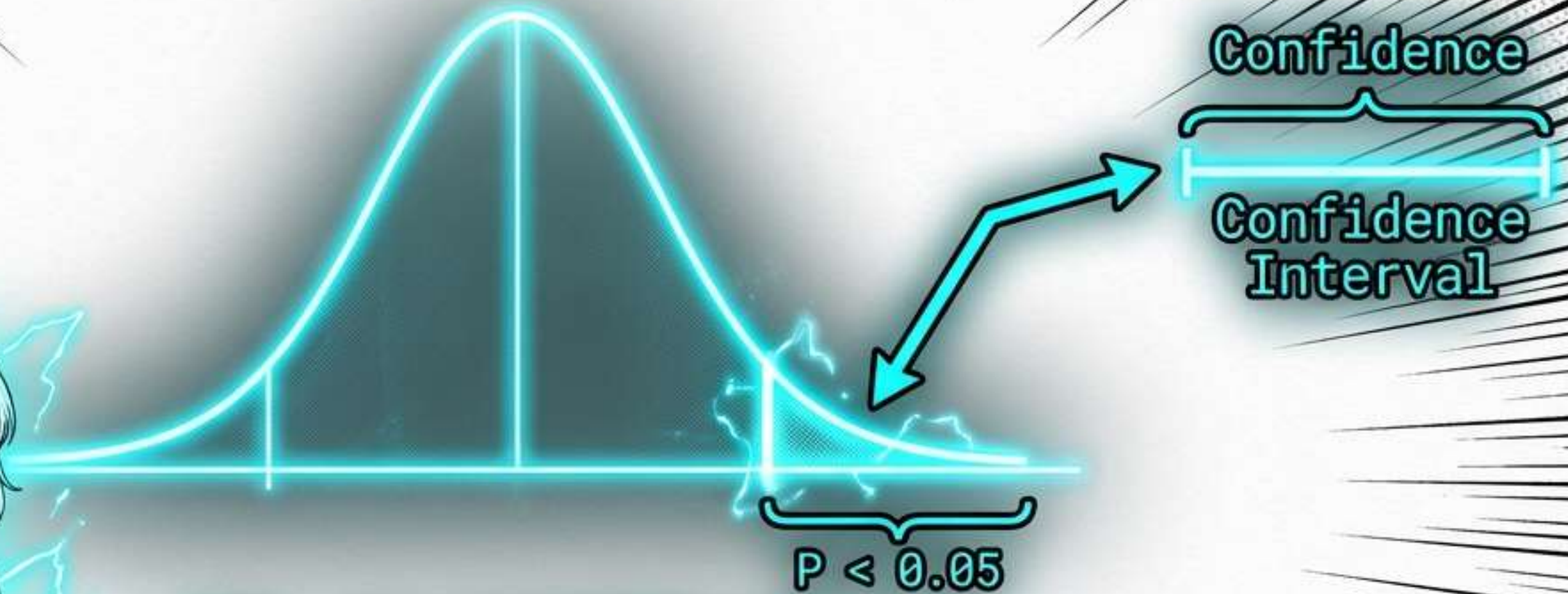
FEATURE	P-VALUE	CONFIDENCE INTERVAL
Nature	Single Value (Point)	Range of Values
Indicates	Statistical Significance	Precision, Significance & Magnitude
Interpretation	Probability under H_0	Plausible range of true effect
Usage	Hypothesis Testing	Estimating Effect Size

THE MATHEMATICAL BOND



We are
two sides...

- They are equivalent methods.
- If the 95% CI excludes the Null Value, then $P < 0.05$.
- The narrower the interval, the smaller the P-value.



...of the
same coin.

COMMON MISCONCEPTIONS



$P > 0.05$ does NOT prove the Null is true! It might just mean we need more data!

- P-Value is NOT the probability that the hypothesis is true!

Statistical Significance does not always mean Practical Significance. A tiny effect can be "Significant" but useless.



THE SUPERIOR STRATEGY.

CIs convey
WHETHER an
effect exists
(like P).

Range

BUT CIs also convey
HOW LARGE the
effect might be.

$P > 0.16 = 1037$

CIs reflect the
degree of **variability**
and **sample size**.

“Confidence intervals should be the preferred means of interpreting results... because in addition to evaluating the role of chance it reflects the degree of variability.”

– Adedokun B.O.

THE RESEARCHER'S GUIDE

- ☒ 1. Check the Design: Is the Null Value 0 (for difference) or 1 (for ratio)?
- ☒ 2. Look for the Range: Does the CI cross the Null Value?
- ☒ 3. Check the Width: Is the estimate precise enough to be useful?
- ☒ 4. Check the P: Is it < 0.05 ? (Use this as a flag, not the whole story).

Ready for action!

Together, they provide the full picture.

